PocketDoctor Mark 5 — Proposed Architecture

Next-Gen Autonomous AI Clinical Triage & Emergency Response System

1. Executive Summary & Overview

PocketDoctor Mark 5 is the next-generation evolution of the autonomous AI medical triage device. Building upon the production breakthroughs of **Mark 4 (v4.8.2 & v4.8.2.3)**, Mark 5 introduces a dual-cloud AI execution pipeline, automated SoftAP captive-portal failover, RFC-compliant SSL SMTP emergency email dispatching, and expanded biometric profile telemetry.

Project Name:	PocketDoctor Mark 5 (Proposed)	**Target Hardware:**	ESP32-S3 Dual-Core 240MHz
Author / Lead:	Vaidik Khurana	**Primary LLM:**	Groq LLaMA 3.3 70B Versatile
Release Path:	tests/beta/	**Fallback LLM:**	Google Gemini 2.5 Flash
Encryption:	TLS 1.3 / SMTPS Port 465	**Networking:**	Wi-Fi 802.11 b/g/n + SoftAP

2. Lessons Learned & Production Fixes from Mark 4 (v4.8.2)

During Mark 4 testing and release validation, key edge cases in embedded Wi-Fi connectivity and SSL SMTP mail transmission were analyzed and resolved. Mark 5 natively incorporates all these critical architectural resolutions:

- **Wi-Fi EEPROM Fallback Validation:** Corrected the EEPROM validation logic to check if (wifiConfigValid && strlen(wifiConfig.ssid) > 0). Resolves lockouts where uninitialized or blank EEPROM records bypassed hardcoded default credentials.
- **Automatic SoftAP Captive Portal (v4.8.2.3):** Introduced automatic Access Point fallback (PocketDoctor-AP at 192.168.4.1). Eliminates web server lockout when station connection fails.
- **RFC 5321 SMTP Payload Compliance (<1,000 Octets/Line):** Formatted all HTML email template strings with explicit \r\n line breaks. Eliminates silent message drops caused by single-line HTML payloads exceeding 1,000 octets.
- **RFC 5322 MIME Email Header Standard:** Added mandatory From:, To:, Reply-To:, and unique Message-ID: headers for 100% spam-filter pass-through.
- **TCP Buffer Cleaning:** Refactored readSmtpResponse() to parse responses line-by-line using readStringUntil('\n'), preventing race conditions from multiline server responses.
- **Gmail Self-Sent Email Behavior Routing:** Documented Gmail's internal routing rule where self-sent emails (Sender == Recipient) bypass the Inbox and land under *Sent / All Mail*.

3. Proposed Mark 5 Engineering Specifications

Subsystem	Mark 4 Specification	Mark 5 Proposed Specification
Microcontroller	ESP32-D0WDQ6 (520KB SRAM)	ESP32-S3-WROOM-1 (8MB PSRAM, Vector AI Accel)
AI Inference Engine	Groq API (LLaMA 3.3 70B)	Dual Engine: Groq LLaMA 3.3 70B (Primary) + Gemini 2.5 Flash (Fallback)
Wi-Fi Provisioning	Station Mode / SoftAP Variant	Auto-Switching Hybrid STA+AP Captive Portal with mDNS (pocketdoctor.local)
Emergency Dispatch	Direct SSL SMTP (Port 465)	Dual-Path: Direct SMTPS + Webhook Push Notifications (Pushover / Twilio)
User Interface	0.96" Monochrome OLED (128x64)	2.4" Color TFT (320x240) + Touch + Dual Analog Joystick
Patient Profiles	3 EEPROM Profiles	10 NVS Profiles with Encrypted Vitals & Intoxication History

4. Repository File Inventory & Release Map

File Path	Version	Description / Role
tests/beta/release/4.8.2.ino	v4.8.2	**Official Main Release.** SSL SMTP engine, EEPROM fix, RFC 5321/5322 compliance, sanitized credentials.
tests/beta/release/4.8.2.3.ino	v4.8.2.3	**SoftAP Variant.** Auto Access Point fallback (PocketDoctor-AP @ 192.168.4.1).
tests/beta/release/4.8.1.ino	v4.8.1	Previous release with basic SOS mode.
tests/beta/release/4.8.ino	v4.8.0	Early v4.8 release with 10-question Groq diagnostic triage.
tests/beta/release/4.7.ino	v4.7.0	Legacy base release (EEPROM magic 0xABCD).
src/main.ino	v4.7.0	Active core source file.
RELEASE_NOTES.md	Docs	Project root Markdown release notes.

Engineering Recommendation: *For immediate hardware deployments requiring zero Wi-Fi setup hassle, flash tests/beta/release/4.8.2.3.ino. For production router-based deployments, flash tests/beta/release/4.8.2.ino.*